

BSc (Hons) Computer Science with Software Engineering - Coventry University
National Institute of Business Management (NIBM)

NB6007CEM - Web API Development

Session 4

Hierarchy

Endpoints



Day 2 (PM) - Build Session

Niranga Dharmaratna - niranga@nibm.lk

S3 Designed the Route Map - S4 Wires It

Nine routes designed in S3. All nine go live this session, serving real data from seed.json.

URI	Resource type
GET /provinces	Collection
GET /provinces/:province-id	Atomic member
GET /districts	Collection
GET /districts/:district-id	Atomic member
GET /stations	Collection
GET /stations/:station-id	Atomic member
GET /vehicles	Collection
GET /vehicles/:vehicle-id	Atomic member
GET /vehicles/:vehicle-id/pings	Scoped collection

Scope today: GET only. No auth. No status codes beyond 200 and 404. Full CRUD arrives in S7-S8.

Read This Before You Generate



**Your generator will return JSON - that is expected and fine for now.
But the field names, nesting depth, and array structure in those response bodies are what the generator chose.
We have not formally decided our representation yet.
Session 5 is where that decision happens.**

Do not get attached to these field names.

The routes you wire today are real design decisions. The response shapes inside them are provisional until S5.

Coursework note: field names accepted at viva are design decisions you must justify. Saying 'the AI generated them' is not a justification. S5 gives you the framework to defend them.

Lab Run Sheet - 180 Minutes

Time	Block	What you do
0-15	Recap	Slides 2-3. Study the 9-route map. Read the representation boundary warning (slide 3) before typing any prompt.
15-45	Generate	Run the prompt scaffold on slide 5 in your AI tool. Read the full output before running node.
45-75	Critique	Work through the §5.1 checklist (slide 6). Check for the rogue /pings route (slide 7). Fix violations before deploying.
75-105	Fix and local test	Correct seed loading if needed (slide 8). Test all 9 endpoints locally with curl or Thunder Client.
105-135	Deploy	Push to your public URL. Confirm all 9 routes return data from seed.json - not hardcoded values.
135-150	Checkpoint	Instructor verifies all 9 endpoints live. Binary pass recorded. Commit to repo.
150-165	Micro-viva	5 questions. Visible on slide 12. You have seen them - answer at your deployed API.
165-180	Buffer / S5 preview	Catch-up and fixes. S5 preview: the representation decision that settles the field-name question from today.

Step 1 - Generate Your Routes

Copy this prompt exactly into your AI tool. Read the full output before running node.

Add Express.js GET routes for the following resources.
Serve data from seed.json loaded into memory at startup.
Route paths must follow REST API Design naming rules:
lowercase, hyphens as separators, plural nouns for collections, singular nouns for members, nouns not verbs.
Do not add a database. Do not add authentication.
Routes to add:
GET /provinces
GET /provinces/:provinceId
GET /districts
GET /districts/:districtId
GET /stations
GET /stations/:stationId
GET /vehicles
GET /vehicles/:vehicleId
GET /vehicles/:vehicleId/pings

Keep the S3 URI-naming linter open in a browser tab. Run the generated paths through it before testing locally.

Step 2 - Critique Before You Run

\$5.1 naming rules - run every generated path through this checklist before starting node

Lowercase only Path segments use lowercase characters only. `/Vehicles` is a violation. `/vehicles` is correct.

Hyphens, not underscores or camelCase `/police_stations` and `/policeStations` are violations. `/police-stations` is correct.

Plural nouns for collections `/province` is a violation. `/provinces` is correct. The plural signals this is a collection resource.

Singular nouns for members `/vehicles/:vehicleId` is correct. A path parameter identifies one member - the collection stays plural.

Nouns, not verbs `GET /getVehicles` is a violation. `GET /vehicles` is correct. The HTTP method carries the verb - the path does not.

The Pings Scoping Rule - §4.6 and §5.6

A rogue GET /pings route is the most common generator mistake on this session. Spot it, remove it.

Generator anti-pattern

```
// Added uninstructed:  
app.get('/pings', (req, res) => {  
  res.json(data.pings);  
});
```

Returns every ping in the system regardless of vehicle. Violates §5.6: the collection of all pings in all vehicles is not of interest - only pings scoped to a specific vehicle are.

Correct scoped form

```
app.get(  
  '/vehicles/:vehicleId/pings',  
  (req, res) => {  
    const p = data.pings.filter(  
      x => x.vehicle_id ===  
        req.params.vehicleId);  
    res.json(p);  
  });
```

Returns pings for this vehicle only. §5.6: pings are a scoped collection - a global /pings route is a design error.

Fix: remove GET /pings entirely - it is a design error, not a bonus route. Micro-viva Q2 tests this directly.

Fix - Load seed.json Once at Startup

If your generator reads the file inside a route handler, move it to startup. This is a checkpoint criterion.

Anti-pattern - reads per request

```
// File read on every request:
app.get('/vehicles', (req, res) => {
  const d = JSON.parse(
    fs.readFileSync(
      './seed.json', 'utf8')
  );
  res.json(d.vehicles);
});
```

Correct - load once at startup

```
// Read once when server starts:
const data = JSON.parse(
  fs.readFileSync(
    './seed.json', 'utf8')
);
app.get('/vehicles', (req, res) => {
  res.json(data.vehicles);
});
```

Critique question: search your generated file for `fs.readFileSync`. How many times does it appear, and where? One call at the top before `app.listen()` is correct. One call per route handler is the anti-pattern.

Micro-viva Q4 tests this: where in your code does seed.json get loaded?

What to Look for in Generated Code

These four failures appear in most AI-generated route files. Catch them in the critique pass before deploying.

Verbs in route names `app.get('/getVehicles')` or `/fetchDistricts`. §5.1: path segments are nouns. GET is already the verb. Fix: rename to `/vehicles` and `/districts`.

Database boilerplate inserted `mongoose.connect()`, `knex`, `pg`, or `Sequelize` models added despite the instruction. Fix: delete the database block entirely. The prompt specified `seed.json` only.

Global GET /pings route A bare `/pings` route returning all pings in the system. §5.6: pings are scoped to a vehicle. Fix: remove it. The only correct form is `/vehicles/:vehicleId/pings`. See slide 7.

seed.json read inside route handlers `fs.readFileSync` inside each `app.get()` callback. Fix: move the read to the top of the file before `app.listen()`. One call, one variable, reused by all routes.

If Routes Fail - Use This Fallback Prompt

Use only if generated routes do not serve seed data after one fix attempt. Copy verbatim.

Generate an Express.js file that:

1. Loads seed.json once at startup using fs.readFileSync.
2. Adds these GET routes, all returning JSON from seed data:
 - GET /provinces - return all provinces
 - GET /provinces/:provinceId - return one province by id
 - GET /districts - return all districts
 - GET /districts/:districtId - return one district by id
 - GET /stations - return all stations
 - GET /stations/:stationId - return one station by id
 - GET /vehicles - return all vehicles
 - GET /vehicles/:vehicleId - return one vehicle by id
 - GET /vehicles/:vehicleId/pings - return pings where vehicle_id matches vehicleId
3. All route paths use lowercase and hyphens only.
4. No database, no auth, no middleware except express.json().

If this also fails: copy the reference build file from the instructor and critique it against §5.1 and the pings scoping rule. Critiquing working code still demonstrates the assessed skill.

Day 2 Checkpoint - Binary Pass

All five must pass before the micro-viva begins. Fail any one - fix and redeploy first.

PASS All 9 routes live and returning seed data GET /provinces, /provinces/:id, /districts, /districts/:id, /stations, /stations/:id, /vehicles, /vehicles/:id, /vehicles/:id/pings - all returning 200 with real seed data.

PASS All route paths pass \$5.1 Confirmed with linter or manual check: lowercase, hyphens, plural collections, no verbs. No /getVehicles. No bare /pings.

PASS /vehicles/:id/pings returns scoped pings only Response contains pings for that vehicle only - not all pings globally. Test with two vehicle IDs and verify the results differ.

PASS seed.json loaded once at startup fs.readFileSync appears at the top of the file before app.listen(). Not inside any route handler.

PASS Committed and deployed to public URL Changes committed to repo. Instructor can reach all 9 endpoints at the public URL - not localhost. Second localhost exception applies only if the host is down.

Micro-Viva - Day 2



You have seen these questions. Answer at your deployed API - not from memory.

Q1 Show me GET /vehicles returning data from your seed. How many vehicles does it return?

Q2 Show me GET /vehicles/{one-of-your-vehicle-ids}/pings. Why is this a scoped collection and not GET /pings?

Q3 Your generator probably produced at least one §5.1 violation. What was it? Which naming rule did it break?

Q4 Where in your code does seed.json get loaded? Why does it matter whether it loads once or per request?

Q5 What is your public URL? Show me any two endpoints returning 200 with real seed data.

Questions & Discussion



Next: S5 brings the representation decision - the field-name question from today gets a formal answer.

Disclaimer & Copyrights



The information presented in this presentation is intended for general informational purposes only. It is not meant to be comprehensive or to constitute professional advice. Any reliance you place on such information is strictly at your own risk. While I strive to keep the information accurate and up-to-date, I make no representations or warranties of any kind, express or implied, about the completeness, accuracy, reliability, suitability, or availability concerning the content within this presentation.

This presentation may include references or links to third-party websites or content. I do not endorse the views expressed or the information provided on such external sites. I am not responsible for the content of any linked site or any link contained in a linked site.

Any action you take upon the information presented in this presentation is strictly at your own discretion. I will not be liable for any losses or damages in connection with the use of this information.

This disclaimer is subject to change without notice.

@ Images and artworks used in this presentation are used for general informational purposes only. It may contain copyrighted material, the use of which may not have been specifically authorised by the copyright owner. This material is available in an effort to explain issues relevant to the topic. This should constitute a 'fair use' of any such copyrighted material.

Last modified: 15 Jun 2026